Caleb Chandler

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EDUCATION

Northeastern University Expected May 2027
M.S., Complex Network Analysis (Concentration: Complex Social Systems)
- GPA: 3.9/4.0

Utah State University Aug 2024
B.S., Economics & Sociology (Dual Major)
- Institutional Certificate of Proficiency - Law and Society

RESEARCH INTERESTS

- *Social Evolution*: The impact of metabolic and information-theoretic constraints on the long-term development of social structure and the scaling relationships observed in human social systems.
- *Collective Computation:* How groups of locally-interacting agents process, distribute, and store information and make inferences about their environment to solve collective problems.
- *Network Science:* Emergence and evolution of structure in real-world networks, how this process can be accurately modeled, and how such models might be used to infer general classes or properties of systems by analyzing the topological histories of their networks.

RESEARCH EXPERIENCE

Graduate Research Assistant Feb 2026 – Present
Social Urban Networks Lab, Northeastern University
PI: Dr. Esteban Moro
Project: Which factors are most responsible for the shape of the visit-dependency network between urban POIs?

- **Built custom link prediction pipeline** in Python from ground up, including mechanism for matching distance distribution of positive and negative edge samples to isolate non-geographic factors
- Independently discovered community structure to be largely redundant with census-block groups, suggesting that shared legal/physical boundaries are reflected in the dependency network's topology
- Constructed “coarse-grained” network by aggregating nodes of individual POIs to tract–category combination nodes, improving distance-controlled link prediction AUC from ~0.59 to ~0.75 and demonstrating that non-geographic factors act on a higher level than that of individual POIs
- **Clearly communicated results and discussed strategy** in weekly meetings

Independent Research Dec 2025 – Present
Network Science Institute, Northeastern University
Advisors: Dr. Iacopo Iacopini and Sagar Kumar
Project: Is command hierarchy a requirement for efficient scaling of collective computation in human societies?

- **Personally owned and oversaw entire process from end-to-end**, including conception, literature review, implementation, writing, and revision.
- **Formulated precise, tractable research design to address literature gap** using DeGroot consensus on a set of model societies varied along two orthogonal axes representing different dimensions of hierarchy
- **Designed and programmed simulation** and analyzed results in Python
- Discovered efficient scaling of coordination cost (consensus time) under each tested condition, contradicting long-standing theory
- Independently sought out faculty advisors and organized weekly meetings to discuss progress and direction

Undergraduate Research Fellow – Energy and Environment Aug 2022 – May 2024
Center for Growth and Opportunity, Utah State University

- Collected data on state-level capacity growth for solar and wind energy and utilized fixed effects regression modeling to support article on disparate market-design schema as potential culprit for differential capacity growth in renewables between states
- Conducted state-by-state writeup of regulatory frameworks and presented findings to senior researcher to assist in development of policy paper calling for clarification of property rights to geothermal resources
- **Independently formulated research project** focused on identifying policy solutions to threat of long-term critical mineral scarcity, diagnosing culprit as mismatched short-vs-long-term incentives and suggesting tax breaks/subsidies for geological surveys
- Assisted Dr. Lucas Rentschler and his team by organizing logistics and recruitment for experimental laboratory studies on game theory and behavioral economics

PROFESSIONAL EXPERIENCE

Data Analyst (Internship) May – Aug 2025
LiveFree Emergency Response

- **Led internal study** comparing quality of leads procured by lead providers, resulting in more profitable lead-acquisition strategy and significant improvement in sales ROI
- Reverse-engineered missing keys to connect calls with customer payment records
- Engineered cohesive data model in Power BI by integrating and transforming large-scale datasets, including calls, payments, and customer records, to uncover initial trends and establish key performance indicators through exploratory analysis
- Efficiently cleaned, validated, and standardized records across multiple unstructured data sources within Power Query
- Wrote Python script to streamline importation of raw customer accounts data into SQL Server

AI Trainer - Statistics Aug 2024 – May 2025
Outlier

- Evaluated performance of LLMs on a range of pre-specified tasks designed to test and improve their ability to interpret statistical data, read charts, graphs and tables, and understand spatial relationships
- Continually strived to improve quality and reliability of work product by adhering to standards and implementing feedback from performance evaluations

PUBLICATIONS & PRESENTATIONS

- **Chandler, C. "**Emergence of command hierarchy not driven by information processing constraints at scale," *Conference on Complex Systems,* Binghamton, NY, USA (Oct 2026). (*Forthcoming*)

ACADEMIC PROJECTS

Financial Contagion in the Between-Country Investment Network Feb - Apr 2026

- Loose replication of Dette et al. (2011) with modern data, examining web of country-level financial entanglements and relative systemic impact of hypothetical collapse/crisis in various countries
- Extended original paper by adapting Battiston et al.'s "DebtRank" algorithm – originally developed for banks – to the inter-country context
- Discovered nested core-periphery structure with vast majority of risk coming from Western countries, as well as an outsized impact from otherwise-minor nations acting as "offshore financial centers"

Network of Events Oct 2025 - Mar 2026

- Collaborative soccer/football analytics project exploring prospect of encoding teams in matches as "event networks," where events (shot, pass, etc) are nodes and links represent chronological succession between them
- Utilized deep learning model to obtain match-level embeddings by "disguising" adjacency matrices as images and processing them through a vision transformer
- Computed distances between embeddings and aggregated them to uncover team signatures

- Created visualizations and presented results to wider research group

College Football Talent Acquisition Network Oct 2025 - Jan 2026

- Network analysis concerning system of player recruitment and transfer in collegiate American football
- Used Python to execute range of network analysis techniques, including community detection, bipartite projection, and bespoke null modeling
- Tracked progress with Git and regularly pushed to public repository
- Wrote algorithm to impute missing attributes, resulting in **~50% improvement in coverage**
- Developed and computed a set of original metrics to represent roster churn and team recruitment strategy
- Communicated results in blog post on personal website

SKILLS & CERTIFICATIONS

Programming & Tools: *Python, Git, Jupyter Notebook, Excel, SQL, Power Query, Power BI, LaTeX*

Analytical: *Graph Theory, Machine Learning, Calculus, Probability and Statistics*

Certifications: *Microsoft Office Specialist: Excel Associate (March 2021)*

LEADERSHIP & ACTIVITIES

- **Vice President**, Filmmaking Club, Utah State University 2022–2024

HONORS & AWARDS

Utah State University

- Huntsman Scholar
- Dean's List (Fall 2021 and 2022)

Other

- National Society of Leadership and Success